

2.7 Logical Arguments

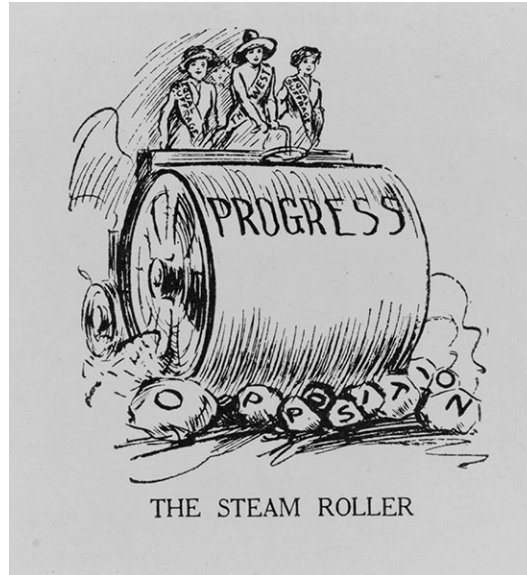


Figure 2.14 Not all logical arguments are valid, and the ongoing fight for equal rights proves that much progress has yet to be made. (credit: "The Steam Roller" by Library of Congress Prints and Photographs Division, public domain)

Learning Objectives

After completing this section, you should be able to:

1. Apply the law of detachment to determine the conclusion of a pair of statements.
2. Apply the law of denying the consequent to determine the conclusion for pairs of statements.
3. Apply the chain rule to determine valid conclusions for pairs of true statements.

The previous sections of this chapter provide the foundational skills for constructing and analyzing logical arguments. All logical arguments include a set of premises that support a claim or conclusion; but not all logical arguments are valid and sound. A logical argument is valid if its conclusion follows from the premises, and it is **sound** if it is valid and all of its premises are true. A false or deceptive argument is called a **fallacy**. Many types of fallacies are so common that they have been named.

?? WHO KNEW?

In 1936, Dale Carnegie published his first book, titled *How to Win Friends and Influence People*. It was marketed as training materials for the improvement of public speaking and negotiation skills, and the methods it presented are still used today. Carnegie famously said, "When dealing with people, remember you are not dealing with creatures of logic, but creatures of emotion."

People who put forth fallacious logical arguments often take advantage of our susceptibility to emotional appeals, to try to convince us that what they are saying is true. The study of logic helps us combat this weakness through recognition and learning to focus on the facts and structure of the argument.

This section focuses on the two main forms that logical arguments can take. While inductive arguments attempt to draw a more general conclusion from a pattern of specific premises, **deductive arguments** attempt to draw specific conclusions from at least one or more general premises. Deductive arguments can be proven to be valid using Venn diagrams or truth tables.

Inductive arguments generally cannot be proven to be true. They are judged as being strong or weak, but, like any opinion, whether you believe an argument is strong or weak often depends on your knowledge of the topic being discussed along with the evidence being provided in the premises. *Hasty generalization* is the name given to any fallacy that presents a weak inductive argument.



Be careful! Premises may be true or false. If a premise is false, the claims made by the argument should be

questioned.

Law of Detachment

The **law of detachment** is a valid form of a conditional argument that asserts that if both the conditional, $p \rightarrow q$, and the hypothesis, p , are true, then the conclusion q must also be true. The law of detachment is also called affirming the hypothesis (or antecedent) and modus ponens. Symbolically, it has the form $((p \rightarrow q) \wedge p) \rightarrow q$.

Law of Detachment	
Premise:	$p \rightarrow q$
Premise:	p
Conclusion:	$\therefore q$

 The $\therefore$ is read as the word, "therefore."

Looking at the truth table for the conditional statement, the only time the conditional is true is when the hypothesis p is also true. The only place this happens is in the first row, where q is also true, confirming that the law of detachment is a valid argument.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Another way to verify that the law of detachment is a valid argument is to construct a truth table for the argument $((p \rightarrow q) \wedge p) \rightarrow q$ and verify that it is a tautology.

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$((p \rightarrow q) \wedge p) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Venn diagrams may also be used to verify deductive arguments, which include conditional premises. Consider the statement $p \rightarrow q$: "If you play guitar, then you are a musician." The set of guitarists is a subset of the set of musicians, $p \subset q$. To verify that an argument is valid using a Venn diagram, draw the Venn diagram representing all the premises in the argument only, as shown in [Figure 2.15](#). Then verify if the conclusion is also represented by the Venn diagram of the premises. If it is, the argument is valid. If it is not, the argument is not valid. The set of guitarists is drawn as a subset of the set of musicians to represent the premise, $p \rightarrow q$. The $\times$ represents the premise: p is true. This completes the drawing of the premises.

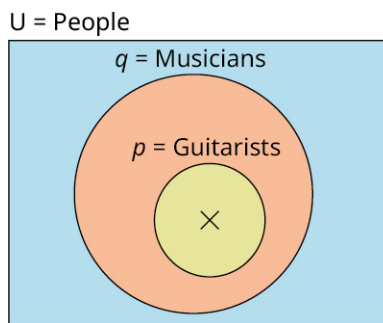


Figure 2.15

Now, examine the Venn diagram to verify if the conclusion is included in the picture. The conclusion is q . Because the x is in the set p , and p is a subset of q , x is also in q ; therefore, the law of detachment is a valid argument.

⚠ Remember that an argument can be valid without being true. For the argument to be proven true, it must be both valid and sound. An argument is sound if all its premises are true.

▶ VIDEO

Logic Part 14: Common Argument Forms like Modus Ponens and Tollens (https://openstax.org/r/Modus_Ponens_and_Tollens)

EXAMPLE 2.31

Applying the Law of Detachment to Determine a Valid Conclusion

Each pair of statements represents the premises in a logical argument. Based on these premises, apply the law of detachment to determine a valid conclusion.

1. If Leonardo da Vinci was an artist, then he painted the *Mona Lisa*. Leonardo da Vinci was an artist.
2. If Michael Jordan played for the Chicago Bulls, then Michael Jordan was not a soccer player. Michael Jordan played for the Chicago Bulls.
3. If all fish have gills, then clown fish have gills. All fish have gills.

✓ Solution

1. The premises are $p \rightarrow q$: If Leonardo da Vinci was an artist, then he painted the *Mona Lisa*, and p : Leonardo da Vinci was an artist. This argument has the form of the law of detachment, so, the conclusion is q : Leonardo da Vinci painted the *Mona Lisa*.
2. The premises follow the form of the law of detachment, so a valid conclusion would be q . The premises are $p \rightarrow q$: If Michael Jordan played for the Chicago Bulls, then Michael Jordan was not a soccer player, and p : Michael Jordan played for the Chicago Bulls. The conclusion that follows from the premises is q : Michael Jordan was not a soccer player.
3. The premises are $p \rightarrow q$: If all fish have gills, then clown fish have gills, and p : All fish have gills. This argument has the form of the law of detachment, so the conclusion is q : clown fish have gills.

> YOUR TURN 2.31

Each pair of statements represents the premises in a logical argument. Based on these premises, apply the law of detachment to determine a valid conclusion.

1. If my classmate likes history, then some people like history. My classmate likes history.
2. If you do not like to read, then some people do not like reading. You do not like to read.
3. If the polygon has five sides, then it is not an octagon. The polygon has five sides.

Law of Denying the Consequent

Another form of a valid conditional argument is called the **law of denying the consequent**, or modus tollens. Recall,

that the conditional statement, $p \rightarrow q$, is logically equivalent to the contrapositive, $\sim q \rightarrow \sim p$. So, if the conditional statement is true, then the contrapositive statement is also true. By the law of detachment, if $\sim q$ is also true, then it follows that $\sim p$ must also be true. Symbolically, it has the form $((p \rightarrow q) \wedge \sim q) \rightarrow \sim p$.

Law of Denying the Consequent	
Premise:	$p \rightarrow q$
Premise:	$\sim q$
Conclusion:	$\therefore \sim p$

 The conditional statement can also be described as, “If antecedent, then consequent.” This is where the law of denying the consequent gets its name.

To verify if the law of denying the consequent is a valid argument, construct a truth table for the argument, $((p \rightarrow q) \wedge \sim q) \rightarrow \sim p$, and verify that it is a tautology.

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$(p \rightarrow q) \wedge \sim q$	$((p \rightarrow q) \wedge \sim q) \rightarrow \sim p$
T	T	F	F	T	F	T
T	F	F	T	F	F	T
F	T	T	F	T	F	T
F	F	T	T	T	T	T

To verify an argument of this form using a Venn diagram, again consider the premise: $p \rightarrow q$: “If you play guitar, then you are a musician.” We will change the second premise to $\sim q$. In this case, the $\times$ represents the premise, $\sim q$. So, it will be placed inside the universal set of all people, but outside the set of musicians, as depicted in the Venn diagram in [Figure 2.16](#).

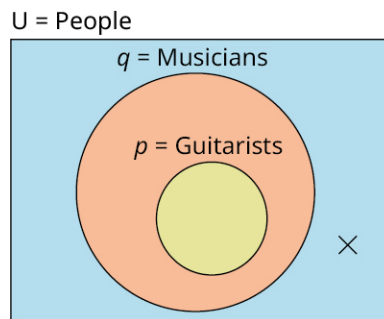


Figure 2.16

Because the $\times$ is also outside the set of guitarists, the statement $\sim p$ follows from the premises and the argument is valid.

EXAMPLE 2.32

Applying the Law of Denying the Consequent to Determine a Valid Conclusion

Each pair of statements represents the premises in a logical argument. Based on these premises, apply the law of denying the consequent to determine a valid conclusion.

1. If Leonardo da Vinci was an artist, then he painted the *Mona Lisa*. Leonardo da Vinci did not paint the *Mona Lisa*.
2. If Michael Jordan played for the Chicago Bulls, then Michael Jordan was not a soccer player. Michael Jordan was a

soccer player.

3. If all fish have gills, then clown fish have gills. Clown fish do not have gills.

Solution

- The premises are $p \rightarrow q$: If Leonardo da Vinci was an artist, then he painted the *Mona Lisa*, and $\sim q$: Leonardo da Vinci did not paint the *Mona Lisa*. This argument has the form of the law of denying the consequent, so the conclusion is $\sim p$: Leonardo da Vinci was not an artist.
- The premises follow the form of the law of denying the consequent, so a valid conclusion would be $\sim p$. The premises are: $p \rightarrow q$: If Michael Jordan played for the Chicago Bulls, then Michael Jordan was not a soccer player, and $\sim q$: Michael Jordan was a soccer player. The conclusion that follows from the premises is $\sim p$: Michael Jordan did not play for the Chicago Bulls.
- The premises are $p \rightarrow q$: If all fish have gills, then clown fish have gills, and $\sim q$: Clown fish do not have gills. This argument has the form of the law denying the consequent, so the conclusion is $\sim p$: Some fish do not have gills.

YOUR TURN 2.32

Each pair of statements represents the premises in a logical argument. Based on these premises, apply the law of denying the consequent to determine a valid conclusion.

- If my classmate likes history, then some people like history. Nobody likes history.
- If Homer does not like to read, then some people do not like reading. All people like reading.
- If the polygon has five sides, then it is not an octagon. The polygon is an octagon.

Chain Rule for Conditional Arguments

The **chain rule for conditional arguments** is another form of a valid conditional argument. It is also called hypothetical syllogism or the transitivity of implication. Recall that the conditional statement $p \rightarrow q$ can also be read as p implies q . This is where the name *transitivity of implication* comes from. The transitive property for numbers states that, if $3 < 4$ and $4 < 5$, then it follows that $3 < 5$. The chain rule extends this property to conditional statements. If the premises of the argument consist of two conditional statements, with the form " $p \rightarrow q$ " and " $q \rightarrow r$," then it follows that $p \rightarrow r$. Symbolically, it has the form $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$.

Chain Rule for Conditional Arguments	
Premise:	$p \rightarrow q$
Premise:	$q \rightarrow r$
Conclusion:	$\therefore p \rightarrow r$

To verify the chain rule for conditional arguments, construct a truth table for the argument, $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$, and verify that it is a tautology.

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

To verify an argument of this form using a Venn diagram, again consider the premise $p \rightarrow q$: “If you play guitar, then you are a musician,” but change the second premise to $q \rightarrow r$: “If you are a musician, then you are an artist.” In this case, the set p of guitarists is a subset of the set r of artists, and it follows that if you are a guitarist, then you are an artist. Therefore, the conclusion $p \rightarrow r$ follows from the premises and the chain rule for logical arguments is valid. See [Figure 2.17](#).

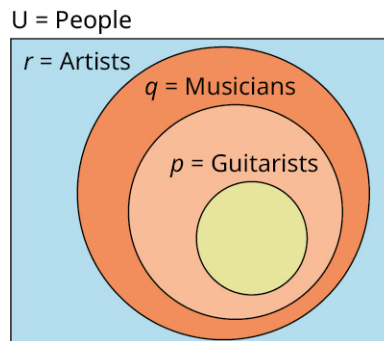


Figure 2.17

EXAMPLE 2.33**Applying the Chain Rule for Conditional Arguments to Determine a Valid and Sound Conclusion**

Each pair of statements represents true premises in a logical argument. Based on these premises, apply the chain rule for conditional arguments to determine a valid and sound conclusion.

1. If my roommate goes to work, then my roommate will get paid. If my roommate gets paid, then my roommate will pay their bills.
2. If robins can fly, then some birds can fly. If some birds can fly, then we will watch birds fly.
3. If Irma is a teacher, then Irma has a college degree. If Irma has a college degree, then Irma graduated from college.

✓ Solution

1. The premises are $p \rightarrow q$: “If my roommate goes to work, then they will get paid,” and $q \rightarrow r$: “If my roommate gets paid, then my roommate will pay their bills.” This argument has the form of the chain rule for conditional arguments, so the valid conclusion will have the form “ $p \rightarrow r$.” Because all the premises are true, the valid and sound conclusion of this argument is: “If my roommate goes to work, then my roommate will pay their bills.”
2. The premises are $p \rightarrow q$: “If robins can fly, then some birds can fly,” and $q \rightarrow r$: “If some birds can fly, then we will watch them fly.” This argument has the form of the chain rule for conditional arguments, so, the valid conclusion will have the form “ $p \rightarrow r$.” Because all the premises are true, the valid and sound conclusion of this argument is: “If robins can fly, then we will watch birds fly.”
3. The premises are $p \rightarrow q$: (see line 1 of solution 1 and 2 above) “If Irma is a teacher, then Irma has a college degree,” and $q \rightarrow r$: “If Irma has a college degree, then Irma graduated from college.” This argument has the form of the chain rule for conditional arguments, so the valid conclusion will have the form “ $p \rightarrow r$.” Because all the premises are true, the valid and sound conclusion of this argument is: “If Irma is a teacher, then Irma graduated from college.”

> YOUR TURN 2.33

Each pair of statements represent true premises in a logical argument. Based on these premises, apply the chain rule for conditional arguments to determine a valid and sound conclusion.

1. If my roommate does not go to work, then my roommate will not get paid. If my roommate does not get paid, then they will not be able to pay their bills.
2. If penguins cannot fly, then some birds cannot fly. If some birds cannot fly, then we will watch the news.
3. If Marcy goes to the movies, then Marcy will buy popcorn. If Marcy buys popcorn, then she will buy water.

Check Your Understanding

34. A _____ is a logical statement used as a fact to support the conclusion of an argument.
35. A logical argument is _____ if its conclusion follows from the premises.
36. A logical argument that attempts to draw a more general conclusion from a pattern of specific premises is called an _____ argument.
37. A _____ argument draws specific conclusions from more general premises.
38. Not all arguments are true. A false or deceptive argument is called a _____.
39. If an argument is valid and all of its premises are true, then it is considered _____.



SECTION 2.7 EXERCISES

For the following exercises, analyze the argument and identify the form of the argument as the law of detachment, the law of denying the consequent, the chain rule for conditional arguments, or none of these.

1. If Apple Inc. releases a new iPhone, then customers will buy it. Customers did not buy a new iPhone. Therefore, Apple Inc. did not release a new iPhone.
2. In the animated movie *Toy Story*, if Paul Newman turned down the role of voicing Woody, then Tom Hanks was chosen for the role. Tom Hanks was chosen as the voice for Woody, therefore, Paul Newman turned down the role of voicing Woody in *Toy Story*.
3. $p \rightarrow q$ and $q \rightarrow r$. $\therefore p \rightarrow r$.
4. $p \rightarrow q$ and p . $\therefore q$.
5. $p \rightarrow q$ and $\sim q$. $\therefore \sim p$.
6. If all people are created equal, then all people are the same with respect to the law. If all people are the same with respect to the law, then justice is blind. Therefore, if all people are created equal, then justice is blind.
7. If I mow the lawn, then my caregiver will pay me twenty dollars. I mowed the lawn. Therefore, my caregiver paid me twenty dollars.
8. If Robin Williams was a comedian, then some comedians are funny. No comedians are funny. Therefore, Robin Williams was not a comedian.

For the following exercises, each pair of statements represents the premises in a logical argument. Based on these premises, apply the law of detachment to determine and write a valid conclusion.

9. $p \rightarrow \sim q$ and p .
10. $\sim p \rightarrow q$ and $\sim p$.
11. If Richard Harris played Dumbledore, then Daniel Radcliffe played Harry Potter. Richard Harris played Dumbledore.
12. If Emma Watson is an actor, then Emma Watson starred as Belle in the movie *Beauty and the Beast*. Emma Watson is an actor.
13. If some Granny Smiths are available, then we will make an apple pie. Some Granny Smiths are available.
14. If Peter Rabbit lost his coat, then all rabbits must avoid Mr. McGregor's garden. Peter Rabbit lost his coat.

For the following exercises, each pair of statements represents the premises in a logical argument. Based on these premises, apply the law of denying the consequent to determine and write a valid conclusion.

15. If Greg and Ralph are friends, then Greg will not play a prank on Ralph. Greg played a prank on Ralph.
16. If Drogon is not a dragon, then Daenerys ruled Westeros. Daenerys did not rule Westeros.
17. $p \rightarrow \sim q$ and q .

18. $\sim p \rightarrow q$ and $\sim q$.
19. If all dragons breathe fire, then rainwings are not dragons. Rainwings are dragons.
20. If some pirates have parrots as pets, then some parrots do not like crackers. All parrots like crackers.

For the following exercises, each pair of statements represent true premises in a logical argument. Based on these premises, apply the chain rule for conditional arguments to determine a valid and sound conclusion.

21. $\sim p \rightarrow q$ and $q \rightarrow \sim r$.
22. $r \rightarrow \sim q$ and $\sim q \rightarrow \sim p$.
23. $q \rightarrow r$ and $p \rightarrow q$.
24. $\sim r \rightarrow p$ and $q \rightarrow \sim r$.
25. If Mr. Spock is a science officer, then Montgomery Scott is an engineer. If Montgomery Scott is an engineer, then James T. Kirk is the captain.
26. If Prince Charles is a character from *Star Wars*, then Luke Skywalker is not a Jedi. If Luke Skywalker is not a Jedi, then Darth Vader is not his father.

For the following exercises, each pair of statements represent true premises in a logical argument. Based on these premises, state a valid conclusion based on the form of the argument.

27. If my siblings drink milk out of the carton, then they will leave the carton on the counter. My siblings did not leave the carton on the counter.
28. If my friend likes to bowl, then my partner does not like to play softball. My friend likes to bowl.
29. If mathematics is fun, then students will study algebra. If students study algebra, then they will score a 100 on their final exam.
30. If all fleas bite and our dog has fleas, then our dog will scratch a lot. Our dog will not scratch a scratch a lot.
31. If the toddler is not tall, then they will use a stepladder to reach the cookie jar. If the toddler will use a stepladder to reach the cookie jar, then they will drop the jar. If they drop the cookie jar, then they will not eat any cookies.
32. If you do not like to dance, then you will not go to the club. You went to the club.

For the following exercises, use a truth table or construct a Venn diagram to prove whether the following arguments are valid.

33. Denying the hypothesis: $p \rightarrow q$ and $\sim p$. Therefore, $\sim q$.
34. Affirming the consequent: $p \rightarrow q$ and q . Therefore, p .
35. $\sim p \vee q$ and p . Therefore, q .
36. $p \wedge q \rightarrow r$ and $\sim r$. Therefore, $\sim p \vee \sim q$.